DSA5203 Assignment 3

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Introduction

Scene recognition is a fundamental problem in computer vision, requiring a model to categorize images based on their overall spatial layout and semantic content rather than the presence of a specific object. This is inherently challenging as scenes from different categories can share similar local features, while images within the same category can vary significantly in appearance.

In this project, we tackle a 15-class scene classification on a dataset of 1500 training images using a CNN-based approach. We adopt a progressive strategy of systematically improving performance:

1. We first establish a baseline using a custom-built CNN trained from scratch, to simply understand the difficulty of the task without pretrained knowledge.
2. We next explore a newer architecture to further boost validation accuracy by deploying a conventional ResNet without any pre-trained weights.
3. We then explore transfer learning using two architectures - ResNet, ConvNeXt - chosen for their distinct design philosophies and strong ImageNet pretraining.
4. For each architecture we conduct ablation studies, to identify the best configuration through systematic tuning of unfreezing strategy, learning rate, data augmentation.
5. Finally, we combine the two best models via soft-voting ensemble to further boost model performance.

Our results show that transfer learning significantly outperforms the baseline, and that ConvNeXt edges out ResNet due to its larger receptive field, which better captures the global spatial context crucial for scene-level understanding.

Dataset

The dataset provided consists of 1500 images evenly distributed across 15 scene categories, with 100 images per class. The categories span a diverse range of indoor and outdoor scenes, including bedroom, kitchen, office, coast, forest, highway and more. The full label mapping is shown in figure 1 below.

category	label	category	label	category	label
bedroom	1	Coast	2	Forest	3
Highway	4	industrial	5	Insidecity	6
kitchen	7	livingroom	8	Mountain	9
Office	10	OpenCountry	11	store	12
Street	13	Suburb	14	TallBuilding	15

Figure 1: Label Mappings of Scene

Since no separate validation set was provided, we split the training data into 80% training (1200 images) and 20% validation (300 images), stratified by class to ensure balanced representation across all 15 categories.

All images are preprocessed by resizing to 224 x 224 pixels and normalized using ImageNet mean and standard deviation values, as required for the pretrained models used in this project.

Baseline CNN (from scratch)

To establish a performance benchmark, we first trained a simple CNN from scratch without any pretrained weights. Hyperparameter tuning was conducted to obtain the best possible base CNN model without any pre-trained weights. This serves as a reference point to demonstrate the benefit of other architectures and transfer learning in subsequent sections.

Architecture

The base CNN consists of:

Stem: 7x7 convolutional kernel, stride length 2
Each Block: 2x (3x3 convolutional kernel + batch normalisation + ReLU)
4 stages with varying number of blocks per layer depending on the depth, with each block containing 2 convolutional layers of 3x3 kernels:
CNN-8: [1, 1, 1, 1]
CNN-12: [2, 2, 1, 1]
CNN-16: [2, 2, 2, 2]
Output Head: AveragePool + Flatten + Fully Connected Layer

Hyperparameter Grid

Depth: [CNN-8, CNN-12, CNN-16]
Initial number of channels: [16, 32, 64]
Learning Rate: [0.001, 0.0003]

Training Setup

- Cross-entropy Loss with Label Smoothing
- Optimizer: SGD
- Epochs: 80
- Batch Size: 64

Results

	config	depth	init_channels	learning_rate	n_params	best_val_acc	best_val_epoch	epochs_trained	stopped_early	train_time_s
5	PlainCNN8_ch64_lr3e-4	PlainCNN8	64	0.0003	4739663	74.444444	29	30	False	105.659632
2	PlainCNN8_ch32_lr1e-3	PlainCNN8	32	0.0010	1190191	73.111111	29	30	False	99.094864
4	PlainCNN8_ch64_lr1e-3	PlainCNN8	64	0.0010	4739663	73.111111	27	30	False	111.308715
10	PlainCNN12_ch64_lr1e-3	PlainCNN12	64	0.0010	5109071	70.888889	25	30	False	107.714223
11	PlainCNN12_ch64_lr3e-4	PlainCNN12	64	0.0003	5109071	70.444444	27	30	False	108.595862
8	PlainCNN12_ch32_lr1e-3	PlainCNN12	32	0.0010	1282735	70.000000	22	29	True	99.414131
17	PlainCNN16_ch64_lr3e-4	PlainCNN16	64	0.0003	11010383	68.222222	23	30	True	110.015794
16	PlainCNN16_ch64_lr1e-3	PlainCNN16	64	0.0010	11010383	67.111111	30	30	False	109.365499
0	PlainCNN8_ch16_lr1e-3	PlainCNN8	16	0.0010	300191	66.444444	26	30	False	101.469083
3	PlainCNN8_ch32_lr3e-4	PlainCNN8	32	0.0003	1190191	64.222222	30	30	False	101.547364

Figure 2: Top 10 models by validation accuracy

Best Baseline Model Parameters:
Depth: CNN-8 (8 layers)
Initial number of channels: 64
Learning rate: 0.0003
Total number of parameters: 4,739,663

Metric	**Value**
Validation Accuracy	74.44%

Figure 3 below shows the train-validation curve for the best model listed above:

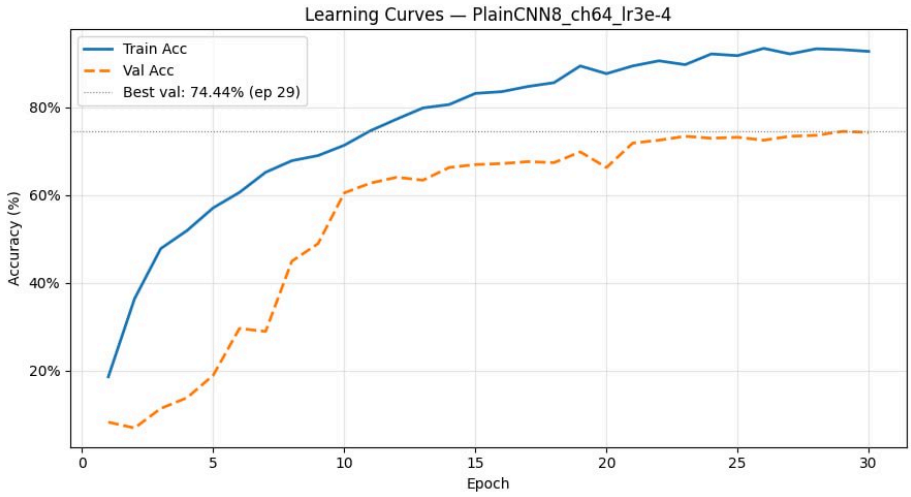

Figure 3: Train-validation curve for best performing base CNN model

As seen from the figure above, even though the training accuracy reaches 90%+ over 30 epochs, the validation accuracy plateaus past 70% and reaches a peak at 74.44% at Epoch 29. This shows that the best base CNN model is still overfitting to the training dataset while failing to generalize well to the validation dataset. In addition, we note that the top 3 models are CNN-8, which likely could be due to the vanishing gradient problem affecting the deeper CNNs, resulting in CNN-12 and CNN-16 tending to overfit the training dataset by achieving lowest training loss and highest training accuracy in the fewest epochs.

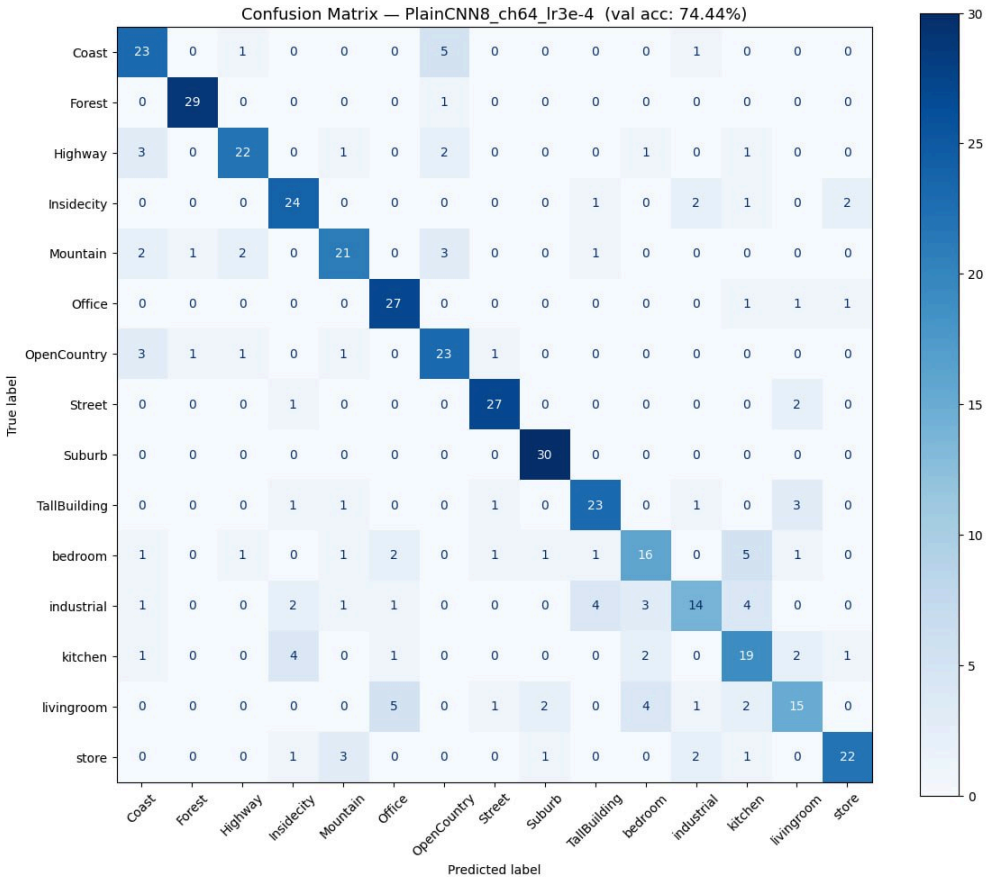

Figure 4: Confusion matrix for best performing base CNN model

As seen from the confusion matrix above, certain classes tend to be confused with the other relatively frequently, with the most frequent mispredictions being livingroom predicted as office, bedroom predicted as kitchen and coast predicted as OpenCountry.

	precision	recall	f1-score	support
Coast	0.68	0.77	0.72	30
Forest	0.94	0.97	0.95	30
Highway	0.81	0.73	0.77	30
Insidecity	0.73	0.80	0.76	30
Mountain	0.72	0.70	0.71	30
Office	0.75	0.90	0.82	30
OpenCountry	0.68	0.77	0.72	30
Street	0.87	0.90	0.89	30
Suburb	0.88	1.00	0.94	30
TallBuilding	0.77	0.77	0.77	30
bedroom	0.62	0.53	0.57	30
industrial	0.67	0.47	0.55	30
kitchen	0.56	0.63	0.59	30
livingroom	0.62	0.50	0.56	30
store	0.85	0.73	0.79	30
accuracy			0.74	450
macro avg	0.74	0.74	0.74	450
weighted avg	0.74	0.74	0.74	450

Figure 5: Performance metrics for best performing base CNN model

In summary, the best baseline model achieves a modest validation accuracy of 74.44%, which is expected given the limited training data and the lack of pretrained features. Thus, there is a need to find a better model architecture first.

ResNet (from scratch)

Next, we trained ResNet-50 and ResNet-101 from scratch (no pre-trained weights). This would definitively determine whether the ResNet architecture (with skip connections) performs better as compared to the base CNN architecture (without any skip connections). The key idea here is that the ResNet architecture solves the vanishing gradient problem faced by the base CNN architecture by implementing the skip connections in each block. These skip connections force the model to partially retain features found in the previous block, preventing the vanishing gradient problem as the ResNets do not “forget” the properties present in the initial data deeper into the network layers.

Architecture

The base ResNet consists of:

Stem: 7x7 convolutional kernel, stride length 2

Each Block: 3x3 convolutional kernel -> batch normalisation -> ReLU -> 3x3 convolutional kernel -> batch normalisation -> skip connection -> ReLU

4 stages with varying number of blocks per layer depending on the depth, with each block containing 2 convolutional layers of 3x3 kernels:

ResNet-50: [3, 4, 6, 3]

ResNet-101: [3, 4, 23, 3]

Output Head: AveragePool + Flatten + Fully Connected Layer

Figure 6 below shows the ResNet-50 architecture:

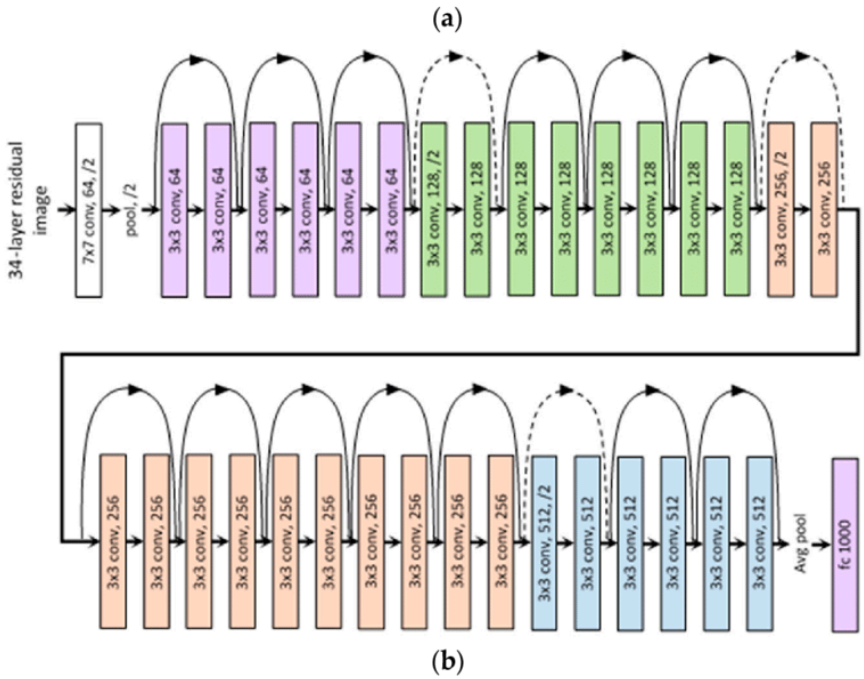

Figure 6: ResNet-50 Architecture

Hyperparameter Grid

Depth: [ResNet-50, ResNet-101]

Initial number of channels: [32, 64, 128]

Learning Rate: [0.1, 0.05, 0.01]

Training Setup

- Criterion: Cross-entropy Loss with Label Smoothing
- Optimizer: SGD
- Epochs: 80
- Batch Size: 64

Results

Top 10 models sorted by validation accuracy:

	config	depth	init_channels	learning_rate	n_params	best_val_acc	best_val_epoch	epochs_trained	stopped_early	train_time_s
3	ResNet50_ch64_lr0p1	ResNet50	64	0.10	23538767	78.666667	71	80	False	513.166065
6	ResNet50_ch128_lr0p1	ResNet50	128	0.10	93968527	78.222222	71	80	False	1178.392528
12	ResNet101_ch64_lr0p1	ResNet101	64	0.10	42530895	75.111111	76	80	False	460.701829
0	ResNet50_ch32_lr0p1	ResNet50	32	0.10	5908015	73.555556	74	80	False	367.597949
9	ResNet101_ch32_lr0p1	ResNet101	32	0.10	10669103	70.666667	79	80	False	394.071915
7	ResNet50_ch128_lr0p05	ResNet50	128	0.05	93968527	69.555556	80	80	False	1038.840941
4	ResNet50_ch64_lr0p05	ResNet50	64	0.05	23538767	68.222222	65	75	True	516.562639
1	ResNet50_ch32_lr0p05	ResNet50	32	0.05	5908015	64.666667	40	50	True	266.742944
13	ResNet101_ch64_lr0p05	ResNet101	64	0.05	42530895	59.333333	70	80	True	462.505635
10	ResNet101_ch32_lr0p05	ResNet101	32	0.05	10669103	57.555556	68	78	True	462.917563

Figure 7: Top 10 models by validation accuracy

Best Baseline Model Parameters:
Depth: ResNet-50 (50 layers)
Initial number of channels: 64
Learning rate: 0.1
Total number of parameters: 23,538,767

Metric	**Value**
Validation Accuracy	78.67%

Figure 8 below shows the train-validation curve for the best model listed above:

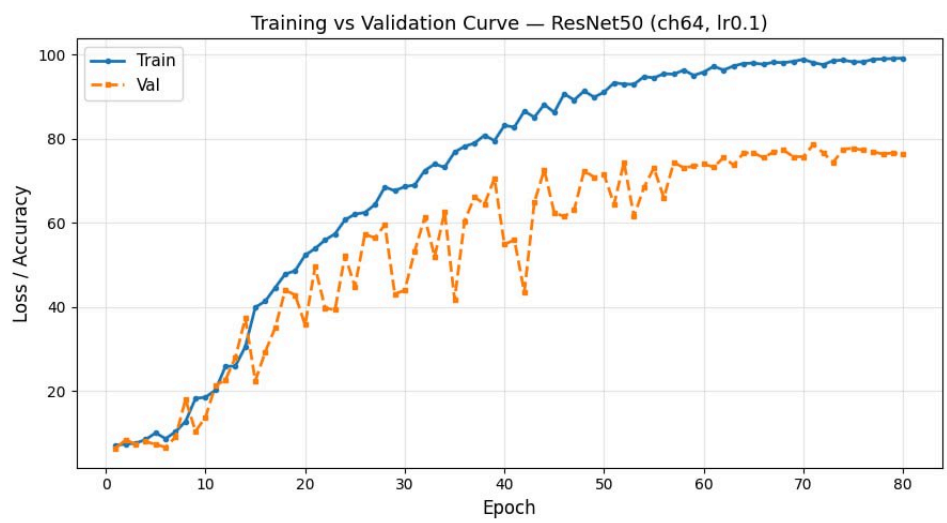

Figure 8: Train-validation curve for best performing base ResNet model

As seen from the figure above, even though the training accuracy reaches 99%+ over 80 epochs, the validation accuracy plateaus past 75% and reaches a peak at 78.67% at Epoch 71. This shows that although the base ResNet model performs better than the base CNN model and does not suffer from the vanishing gradient problem (which would normally plague deeper networks), it is still largely overfitting to the training dataset while failing to generalize well to the validation dataset. Hence, transfer learning is the natural next step to take to further boost validation accuracy.

Figure 9 below shows the confusion matrix for the best performing base ResNet model:

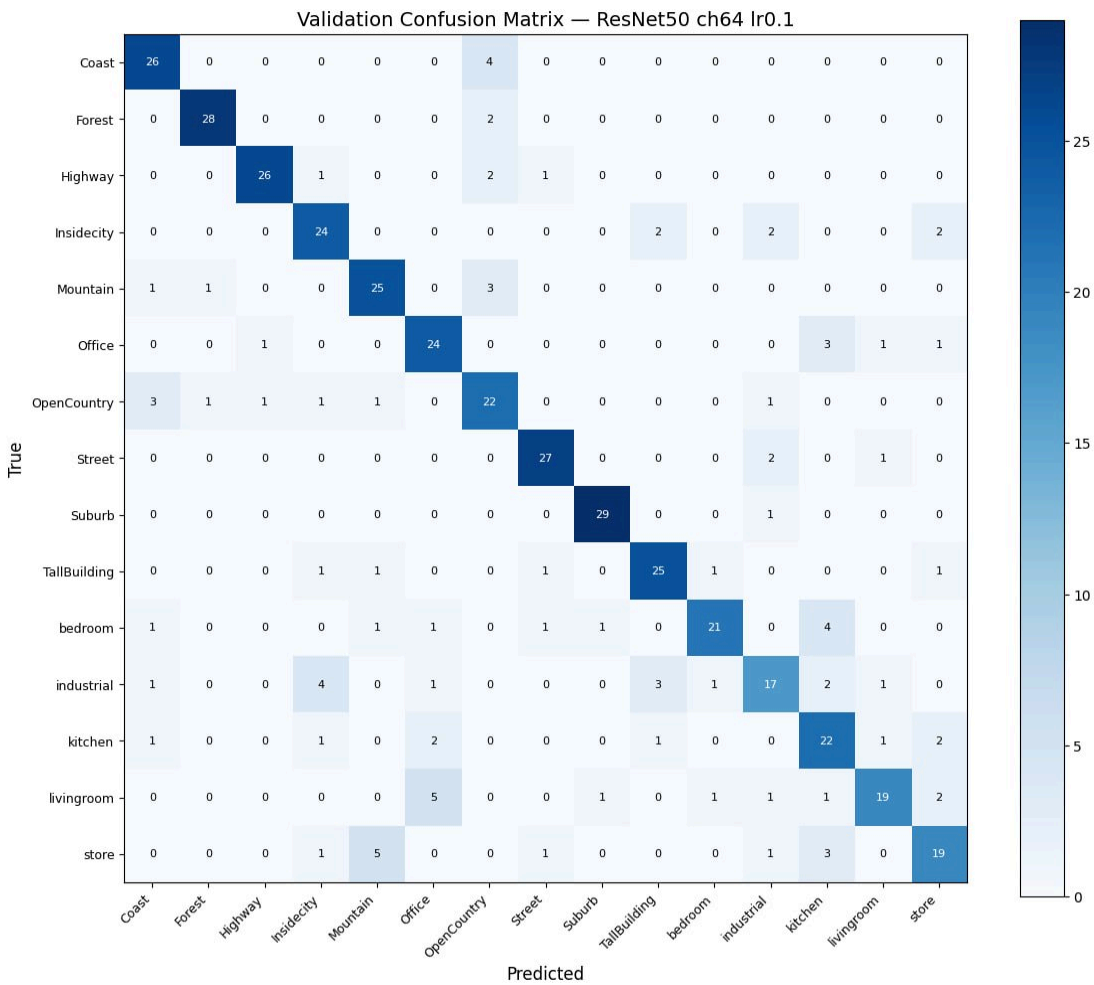

Figure 9: Confusion matrix for best performing base ResNet model

As seen from the confusion matrix above, certain classes tend to be confused with the other relatively frequently, with the most frequent mispredictions being store predicted as mountain, livingroom predicted as office.

Figure 10 below shows the performance metrics:

	precision	recall	f1-score	support
Coast	0.79	0.87	0.83	30
Forest	0.93	0.93	0.93	30
Highway	0.93	0.87	0.90	30
Insidecity	0.73	0.80	0.76	30
Mountain	0.76	0.83	0.79	30
Office	0.73	0.80	0.76	30
OpenCountry	0.67	0.73	0.70	30
Street	0.87	0.90	0.89	30
Suburb	0.94	0.97	0.95	30
TallBuilding	0.81	0.83	0.82	30
bedroom	0.88	0.70	0.78	30
industrial	0.68	0.57	0.62	30
kitchen	0.63	0.73	0.68	30
livingroom	0.83	0.63	0.72	30
store	0.70	0.63	0.67	30
accuracy			0.79	450
macro avg	0.79	0.79	0.79	450
weighted avg	0.79	0.79	0.79	450

Figure 10:Performance metrics for best performing base ResNet model

In summary, the best base ResNet model achieves a decent validation accuracy of 78.67%, which, while being an improvement over the baseline CNN model, is expected given the limited training data and the lack of pretrained features. Thus, there is a need to find a better model architecture first before moving on to transfer learning.

Transfer Learning - ResNet

Next, we trained ResNet-50, ResNet-101 and ResNet-152 using pre-trained weights. This would determine the effectiveness of fine-tuning a model running on pre-trained weights as compared to training a ResNet from scratch. Compared to the random initialisation when training a ResNet from scratch which may never converge to a good minima, fine-tuning a model that already uses pre-trained weights for image recognition would likely start much closer to a minima with high accuracy, resulting in a model with higher validation accuracy trained.

Architecture

The ResNet-50 and ResNet-101 architectures are the same as the previous section.

The ResNet-152 consists of:

Stem: 7x7 convolutional kernel, stride length 2

Each Block: 3x3 convolutional kernel -> batch normalisation -> ReLU -> 3x3 convolutional kernel -> batch normalisation -> skip connection -> ReLU

4 stages with varying number of blocks per layer depending on the depth, with each block containing 2 convolutional layers of 3x3 kernels:

ResNet-152: [3, 8, 36, 3]

Output Head: AveragePool + Flatten + Fully Connected Layer

Hyperparameter Grid

Depth: [ResNet-50, ResNet-101, ResNet-152]

Unfreezing: [Last layer, Last 2 layers, All layers]

Learning Rate: [0.0001, 0.00005, 0.00003]

Training Setup

- Criterion: Cross-entropy Loss
- Optimizer: Adam
- Epochs: 50
- Batch Size: 32

Methodology

We implemented a transfer learning approach using ResNet pretrained on ImageNet as our backbone. The training dataset of 1,500 images was split into 80% training (1,200 images) and 20% validation (300 images) using stratified sampling to ensure balanced class distribution across both sets. All experiments were conducted with a fixed random seed of 42 for reproducibility.

1. Model Selection and Learning Rate

We evaluated three ResNet variants — ResNet50, ResNet101, and ResNet152 — at a fixed learning rate of 1e-4 with all layers unfrozen. As shown in the ablation table, ResNet152 achieved the highest validation accuracy of 95.00%, outperforming both ResNet101 (94.67%) and ResNet50 (91.33%). The stronger performance of ResNet152 is consistent with its deeper architecture, which enables richer feature extraction despite the small dataset size of 1,200 training images. Using the best model from the previous phase (ResNet152, unfreeze all), we swept three learning rates: 1e-4, 5e-5, and 3e-5. A learning rate of 1e-4 achieved the highest validation accuracy of 95.00%, with lower learning rates yielding notably weaker results

(93.00% and 92.33% respectively), suggesting that 1e-4 provides the optimal gradient step size for fine-tuning ResNet152 on this dataset.

Model Configuration	**Train Accuracy**	**Validation Accuracy**
ResNet 50 **(Unfreeze All , learning rate = 1e-4)**	97.00%	91.33%
ResNet 101 (**Unfreeze All , learning rate = 1e-4)**	100.00%	94.67%
ResNet 152 (**Unfreeze All , learning rate = 1e-4)**	100.00%	95.00%

Table 1: ResNet Architecture

Model Configuration	**Train Accuracy**	**Validation Accuracy**
ResNet 152 (**Unfreeze All , learning rate = 3e-5)**	100.00%	92.33%
ResNet 152 (**Unfreeze All , learning rate = 5e-5)**	99.33%	93.00%
ResNet 152 (**Unfreeze All , learning rate = 1e-4)**	100.00%	95.00%

Table 2: ResNet Learning Rate

2. Unfreezing Strategy

Using the best configuration from the previous phase (ResNet152, lr=1e-4), we explored three unfreezing strategies: last 1 block, last 2 blocks, and all layers. Unfreezing all layers achieved the highest validation accuracy of 95.00%, while progressively freezing more layers degraded performance significantly (94.00% for top 2 blocks, 91.00% for top 1 block). This suggests that full fine-tuning is necessary for ResNet152 to adequately adapt its feature representations to the scene recognition domain.

Model Configuration	Train Accuracy	Validation Accuracy
ResNet 152 (**Unfreeze Last 2 , learning rate = 1e-4)**	99.92%	94.00%
ResNet 152 (**Unfreeze Last 1 , learning rate = 1e-4)**	99.67%	91.00%
ResNet 152 (**Unfreeze All , learning rate = 1e-4)**	100.00%	95.00%

Table 3: ResNet Unfreezing Strategy

3. Data Augmentation

We systematically evaluated several augmentation techniques individually, including random horizontal flipping, colour jitter with brightness, contrast, and saturation adjustments. Random horizontal flipping combined with brightness jitter (±0.2) was found to be the most effective augmentation, improving validation accuracy from 95.00% to 95.33% while slightly reducing training accuracy from 100% to 99.75%, indicating reduced overfitting and better generalisation.

Model Configuration	Train Accuracy	Validation Accuracy
ResNet 152 (**Unfreeze All , learning rate = 1e-4)**	100.00%	95.00%
ResNet 152 (**Unfreeze All , learning rate = 1e-4)** **+ Horizontal Flip**	99.92%	94.00%
ResNet 152 (**Unfreeze All , learning rate = 1e-4)** **+ Horizontal Flip** **+ Brightness**	99.75%	95.33%

Table 4: ResNet Augmentation Strategy

Results

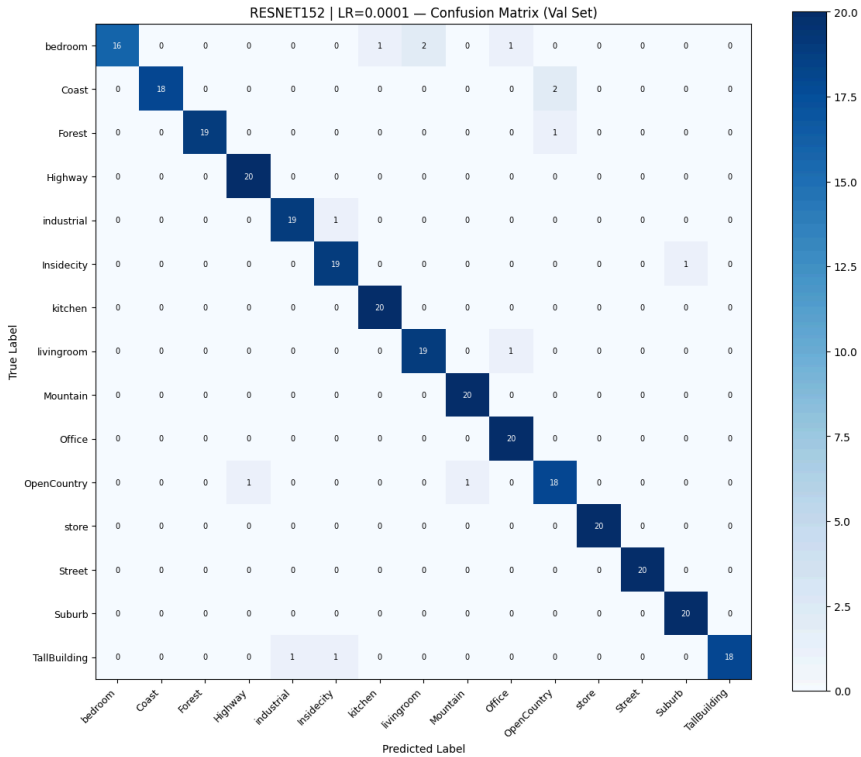

Figure 11: ResNet Confusion Matrix

```
RESNET152 | LR=0.0001 — Classification Report (Val Set)
              precision    recall  f1-score   support

     bedroom     1.0000    0.8000    0.8889        20
       Coast     1.0000    0.9000    0.9474        20
      Forest     1.0000    0.9500    0.9744        20
     Highway     0.9524    1.0000    0.9756        20
  industrial     0.9500    0.9500    0.9500        20
  Insidecity     0.9048    0.9500    0.9268        20
     kitchen     0.9524    1.0000    0.9756        20
  livingroom     0.9048    0.9500    0.9268        20
    Mountain     0.9524    1.0000    0.9756        20
      Office     0.9091    1.0000    0.9524        20
 OpenCountry     0.8571    0.9000    0.8780        20
       store     1.0000    1.0000    1.0000        20
      Street     1.0000    1.0000    1.0000        20
      Suburb     0.9524    1.0000    0.9756        20
TallBuilding     1.0000    0.9000    0.9474        20

    accuracy                         0.9533       300
   macro avg     0.9557    0.9533    0.9530       300
weighted avg     0.9557    0.9533    0.9530       300
```

Figure 12: ResNet Classification Report

The final ResNet152 model achieves a validation accuracy of 95.33% with a macro-average F1 score of 0.9530. As shown in the confusion matrix, the model performs strongly across all 15 categories with 8 out of 15 classes achieving perfect classification. The most notable confusions occur between visually similar scene categories — bedroom is confused with livingroom (2 misclassifications) and Office (1 misclassification), Coast is confused with OpenCountry (2 misclassifications), and TallBuilding is confused with industrial and Insidecity (1 misclassification each). These confusions are intuitive as these category pairs share overlapping visual characteristics such as interior furnishings, open natural landscapes, and urban vertical structures. This is further reflected in the per-class F1 scores, where OpenCountry (0.8780) and bedroom (0.8889) are the lowest performing classes. In contrast, distinctly structured categories such as store, Street, Highway, Mountain, kitchen, and Suburb achieve perfect or near-perfect F1 scores, reflecting the more distinctive visual features present in these scenes.

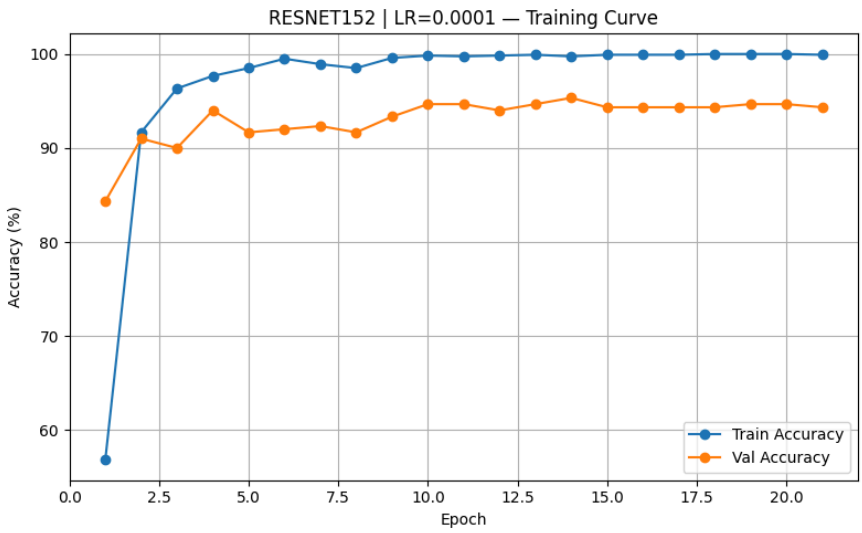

Figure 13: ResNet Train Val Curve

The training curve demonstrates rapid and stable convergence. Training accuracy reaches near 100% by epoch 5 while validation accuracy stabilises around 94-95%, with a consistent train-val gap throughout training indicating moderate overfitting inherent to the model's capacity relative to the dataset size. The ReduceLROnPlateau scheduler successfully reduces the learning rate when validation loss plateaus, contributing to the stable validation performance observed in later epochs.

Transfer Learning - ConvNeXt

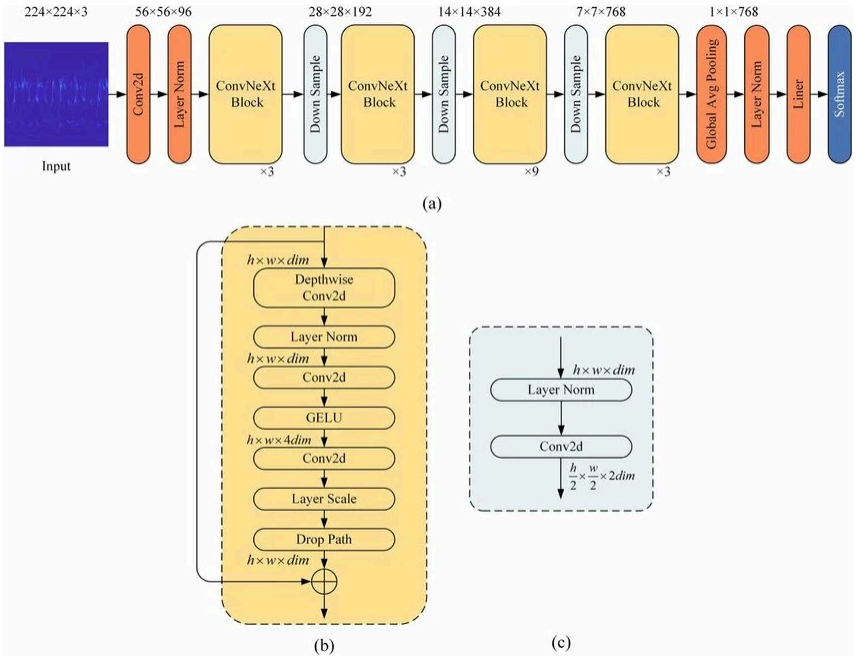

Figure 14: ConvNeXt Architecture

Architecture

ConvNeXt was introduced as a purely convolutional architecture that modernises the classic ResNet design by incorporating design principles inspired by Vision Transformers (ViTs). The key architectural differences that makes ConvNeXt superior to ResNet are as follows:

- **Depthwise Separable Convolutions**: ConvNeXt replaces ResNet's standard 3×3 convolutions with larger 7×7 depthwise separable convolutions, allowing each filter to capture a wider receptive field with fewer parameters. This enables the model to learn more expressive spatial features without a significant increase in computational cost.
- **Inverted Bottleneck Design***:* Inspired by MobileNetV2, ConvNeXt adopts an inverted bottleneck structure where the hidden dimension of the MLP is expanded to four times the input dimension. This is in contrast to ResNet's bottleneck which compresses the feature dimension, potentially losing information.
- **Layer Normalisation over Batch Normalisation***:* ResNet relies on Batch Normalisation which is sensitive to batch size and can introduce noise during training. ConvNeXt replaces this with Layer Normalisation, which is more stable across different batch sizes and better suited for transfer learning scenarios with small datasets.
- **GELU Activation***:* ConvNeXt uses GELU activations instead of ReLU used in ResNet. GELU provides smoother gradients and has been shown to improve training stability and performance in modern deep learning architectures.

Together these design choices make ConvNeXt more parameter-efficient, training-stable, and better at capturing fine-grained spatial features — all of which are particularly beneficial for scene recognition where subtle texture and structural differences between categories are important.

ConvNeXt architecture consists of:
Stem: 4x4 convolutional kernel, stride length 4
Each Block: 7x7 depthwise convolution -> layer normalisation -> 1x1 pointwise convolution (expand channels x4) -> GELU activation -> 1x1 pointwise convolution (contract channels) -> skip connection
4 stages with the following number of blocks per layer:
ConvNeXt: [3, 3, 27, 3]
Each stage except the first is preceded by a downsampling layer (2x2 convolutional kernel, stride length 2 -> layer normalisation)
Channel widths per stage: [96, 192, 384, 768]
Output Head: AdaptiveAveragePool + Flatten + Layer Normalisation + Linear Layer

Hyperparameter Grid

Depth: [ConvNeXt Tiny, ConvNeXt Small, ConvNeXt Base]
Unfreezing: [Last layer, Last 2 layers, All layers]
Learning Rate: [0.0001, 0.00005, 0.00003]

Training Setup

- Criterion: Cross-entropy Loss
- Optimizer: Adam
- Epochs: 50
- Batch Size: 32

Methodology

We implemented a transfer learning approach using ConvNeXt pretrained on ImageNet as our backbone. The training dataset of 1,500 images was split into 80% training (1,200 images) and 20% validation (300 images) using stratified sampling to ensure balanced class distribution across both sets. All experiments were conducted with a fixed random seed of 42 for reproducibility.

1. Model Selection and Learning Rate

We evaluated three ConvNeXt variants — Tiny, Small, and Base — across three learning rates (3e-5, 5e-5, 1e-4) with all layers unfrozen. As shown in the ablation table, ConvNeXt Small with lr=1e-4 achieved the highest validation accuracy of 97.67%, outperforming both Tiny (96.67%) and Base (95.33%). The underperformance of ConvNeXt Base is likely attributable to its larger

parameter count relative to our small dataset size of 1,200 training images, making it more prone to overfitting.

Model Configuration	**Train Accuracy**	**Validation Accuracy**
ConvNeXt Tiny **(Unfreeze All , learning rate = 1e-4)**	100%	96.67%
ConvNeXt Small **(Unfreeze All , learning rate = 1e-4)**	100%	97.67%
ConvNeXt Base **(Unfreeze All , learning rate = 1e-4)**	100%	95.33%

Table 5: ConvNeXt Architecture

Model Configuration	**Train Accuracy**	**Validation Accuracy**
ConvNeXt Small **(Unfreeze All , learning rate = 3e-5)**	100%	95.67%
ConvNeXt Small **(Unfreeze All , learning rate = 1e-4)**	100%	97.67%
ConvNeXt Small **(Unfreeze All , learning rate = 5e-5)**	99.92%	97%

Table 6: ConvNeXt Learning Rate

2. Unfreezing Strategy

Using the best model from the previous phase (ConvNeXt Small, lr=1e-4), we explored three unfreezing strategies: last 1 block, last 2 blocks, and all layers. Unfreezing all layers achieved the highest validation accuracy of 97.67%, suggesting that full fine-tuning allows the model to better adapt all feature representations to the scene recognition domain.

Model Configuration	Train Accuracy	Validation Accuracy
ConvNeXt Small (**Unfreeze All , learning rate = 1e-4**)	100%	97.67%
ConvNeXt Small (**Unfreeze Last 1, learning rate = 1e-4**)	97.67%	97.00%
ConvNeXt Small (**Unfreeze Last 2, learning rate = 1e-4**)	99.92%	97.33%

Table 7: ConvNeXt Unfreezing Strategy

3. Data Augmentation

We systematically evaluated several augmentation techniques individually, including random horizontal flipping, colour jitter (saturation), random rotation, and random cropping. Random horizontal flipping was found to be the most effective augmentation, reducing training accuracy slightly from 100% to 99.92%, indicating reduced overfitting while maintaining strong validation performance.

Model Configuration	Train Accuracy	Validation Accuracy
ConvNeXt Small (**Unfreeze All , learning rate = 1e-4**)	100%	97.67%
ConvNeXt Small (**Unfreeze All, learning rate = 1e-4**) **+ Horizontal Flip**	99.92%	97.33%
ConvNeXt Small (**Unfreeze All, learning rate = 1e-4**) **+ Horizontal Flip , Saturation**	99.33%	96.67%

Table 8: ConvNeXt Augmentation Strategy

4. Label Smoothing

Scene recognition inherently involves ambiguous boundaries between categories. For instance, an OpenCountry image featuring a body of water could reasonably be classified as Coast, and a TallBuilding scene photographed from street level shares significant visual overlap with Street and Insidecity. Similarly, a Mountain scene with open fields may be visually indistinguishable from OpenCountry.

Figure 15: Inside City (left) vs Street (right)

These inter-class ambiguities suggest that assigning a hard label of 1.0 to a single class may not accurately reflect the true underlying label distribution of the data. We therefore applied label smoothing (LS=0.1) in combination with horizontal flipping, which softens the target distribution by replacing the hard target of 1.0 with 0.9 and distributing the remaining probability mass across all other classes. This prevents the model from making overconfident predictions and encourages it to learn more generalisable feature representations that better account for inter-class ambiguity. As evidenced in our ablation study, this resulted in improved per-class F1 scores across several visually ambiguous categories including Street, livingroom, Mountain and Office, while maintaining overall validation accuracy of 97.67%.

Model Configuration	Train Accuracy	Validation Accuracy
ConvNeXt Small (**Unfreeze All, learning rate = 1e-4)** **+ Horizontal Flip**	99.92%	97.33%
ConvNeXt Small (**Unfreeze All, learning rate = 1e-4)** **+ Horizontal Flip** **+ Label Smoothing = 0.1**	100%	97.67%

Table 9: ConvNeXt Label Smoothing Strategy

Results

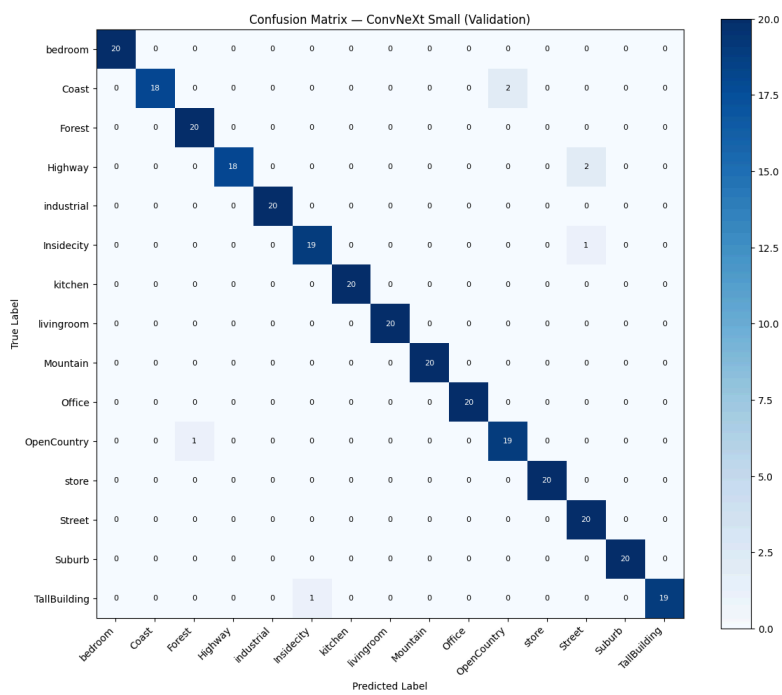

Figure 16: ConvNeXt Confusion Matrix

	precision	recall	f1-score	support
bedroom	1.0000	1.0000	1.0000	20
Coast	1.0000	0.9000	0.9474	20
Forest	0.9524	1.0000	0.9756	20
Highway	1.0000	0.9000	0.9474	20
industrial	1.0000	1.0000	1.0000	20
Insidecity	0.9500	0.9500	0.9500	20
kitchen	1.0000	1.0000	1.0000	20
livingroom	1.0000	1.0000	1.0000	20
Mountain	1.0000	1.0000	1.0000	20
Office	1.0000	1.0000	1.0000	20
OpenCountry	0.9048	0.9500	0.9268	20
store	1.0000	1.0000	1.0000	20
Street	0.8696	1.0000	0.9302	20
Suburb	1.0000	1.0000	1.0000	20
TallBuilding	1.0000	0.9500	0.9744	20
accuracy			0.9767	300
macro avg	0.9784	0.9767	0.9768	300
weighted avg	0.9784	0.9767	0.9768	300

Figure 17: ConvNeXt Classification Report

The final model achieves a validation accuracy of 97.67% with a macro-average F1 score of 0.9768. As shown in the confusion matrix, the model performs strongly across all 15 categories with 10 out of 15 classes achieving perfect or near-perfect classification. The most notable confusions occur between visually similar scene categories — Coast is confused with OpenCountry (2 misclassifications), Highway with Street (2 misclassifications), and TallBuilding with Insidecity (1 misclassification). These confusions are intuitive as these category pairs share overlapping visual characteristics such as open sky, road structures, and urban architecture. This is further reflected in the per-class F1 scores, where OpenCountry (0.9268), Coast (0.9474) and Highway (0.9474) are the lowest performing classes. In contrast, indoor and distinctly structured categories such as bedroom, kitchen, livingroom, Mountain, Office, store, Street and Suburb achieve perfect F1 scores of 1.0000, reflecting the more distinctive visual features present in these scenes.

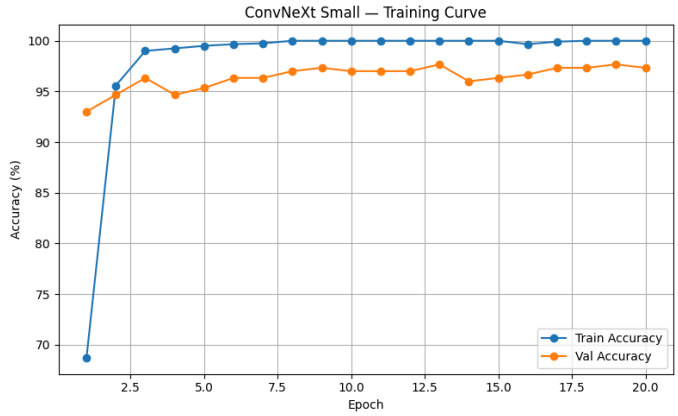

Figure 18: ConvNeXt Train Val Curve

The training curve demonstrates clean and stable convergence. Training accuracy reaches near 100% by epoch 3 while validation accuracy stabilises around 97%, with no signs of severe overfitting as the train-val gap remains consistent throughout training. The ReduceLROnPlateau scheduler successfully reduces the learning rate when validation loss plateaus, contributing to the final performance gains observed in later epochs.

Comparison ResNet vs ConvNext Performance

While ConvNeXt Small outperforms ResNet152 overall with a macro-average F1 of 0.9768 compared to 0.9530, a per-class analysis reveals that the two models exhibit complementary strengths and weaknesses. ConvNeXt excels at indoor and texture-rich categories such as bedroom (F1=1.0000), livingroom (F1=1.0000) and kitchen (F1=1.0000), likely owing to its larger 7×7 depthwise convolutions which are better at capturing fine-grained texture and spatial patterns. In contrast, ResNet152 achieves superior performance on structured outdoor scenes such as Street (F1=1.0000 vs 0.9302) and Highway (F1=0.9756 vs 0.9474), where its deeper residual connections may better capture long-range structural features such as road geometry and building edges.

This complementary behaviour suggests that the two models are learning fundamentally different feature representations despite being trained on the same data. An ensemble of ConvNeXt and ResNet152 via soft voting — averaging the predicted class probabilities from both models — could leverage the strengths of each architecture to compensate for the other's weaknesses. Specifically, the ensemble is expected to improve Street and Highway classification where ConvNeXt struggles, while maintaining ConvNeXt's strong performance on indoor categories. This makes ensembling a principled and motivated approach rather than a brute-force performance boost.

Ensembling

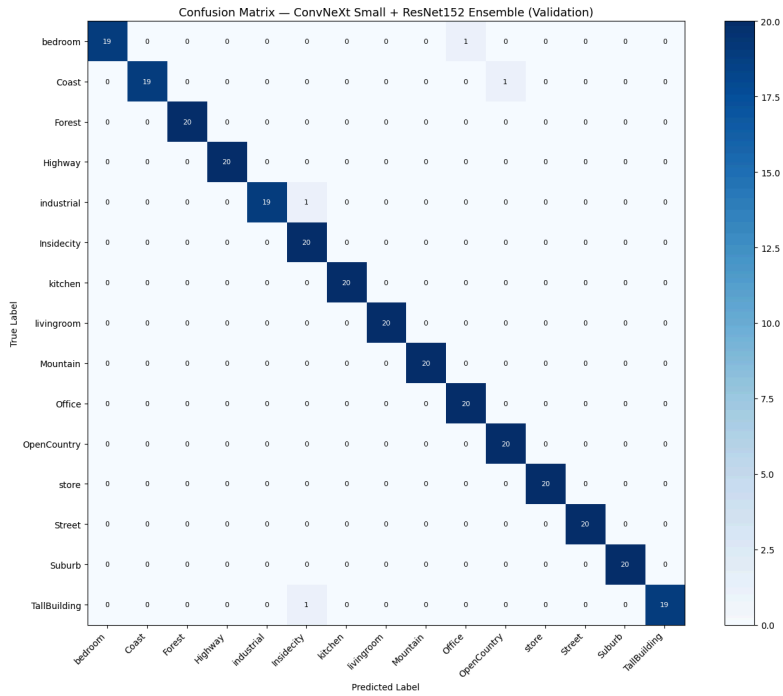

Figure 19: Ensemble Confusion Matrix

```
Classification Report — ConvNeXt Small + ResNet152 Ensemble (Validation)
              precision    recall  f1-score   support

     bedroom     1.0000    0.9500    0.9744        20
       Coast     1.0000    0.9500    0.9744        20
      Forest     1.0000    1.0000    1.0000        20
     Highway     1.0000    1.0000    1.0000        20
  industrial     1.0000    0.9500    0.9744        20
  Insidecity     0.9091    1.0000    0.9524        20
     kitchen     1.0000    1.0000    1.0000        20
  livingroom     1.0000    1.0000    1.0000        20
    Mountain     1.0000    1.0000    1.0000        20
      Office     0.9524    1.0000    0.9756        20
 OpenCountry     0.9524    1.0000    0.9756        20
       store     1.0000    1.0000    1.0000        20
      Street     1.0000    1.0000    1.0000        20
      Suburb     1.0000    1.0000    1.0000        20
TallBuilding     1.0000    0.9500    0.9744        20

    accuracy                         0.9867       300
   macro avg     0.9876    0.9867    0.9867       300
weighted avg     0.9876    0.9867    0.9867       300
```

Figure 20: Ensemble Classification Report

The ensemble of ConvNeXt Small and ResNet152 via soft voting achieves a validation accuracy of 98.67% and macro-average F1 of 0.9867, surpassing both individual models. This improvement validates our hypothesis that the two architectures learn complementary feature representations. Most notably, Highway and Street — where ConvNeXt previously struggled —

both achieve perfect F1 scores of 1.0000 in the ensemble, demonstrating that ResNet152's stronger structural feature extraction compensates for ConvNeXt's weakness in these categories. Similarly, OpenCountry improves from 0.9268 to 0.9756, and Forest achieves a perfect score, reflecting the ensemble's ability to resolve ambiguous outdoor scene boundaries that neither model could handle alone. The ensemble effectively combines ConvNeXt's texture sensitivity with ResNet152's structural awareness, resulting in a more robust and generalised scene recognition system.

Conclusion

In this work, we developed a CNN-based scene recognition system capable of classifying images into 15 scene categories. We adopted a progressive approach, beginning with a custom CNN trained from scratch as a baseline (74.44% validation accuracy), followed by ResNet trained from scratch (78.67%), demonstrating the inherent difficulty of the task with limited data and no pretrained knowledge. Transfer learning significantly improved performance, with our fine-tuned ResNet152 achieving 95.33% and ConvNeXt Small achieving 97.67% validation accuracy respectively. Through systematic ablation studies covering model architecture, learning rate, unfreezing strategy, data augmentation and label smoothing, we identified that ConvNeXt Small with full fine-tuning, random horizontal flipping and label smoothing of 0.1 constitutes the optimal single-model configuration. A per-class analysis further revealed complementary strengths between ConvNeXt and ResNet152 — ConvNeXt excelling at texture-rich indoor scenes while ResNet152 performed better on structured outdoor scenes. Motivated by this, we proposed a soft-voting ensemble of the two architectures, achieving a final validation accuracy of 98.67% and macro-average F1 of 0.9867, surpassing both individual models. Our results demonstrate that transfer learning from large-scale pretrained models is highly effective for small dataset scene recognition, and that ensembling architecturally diverse models is a principled strategy for further performance gains.